

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 3: Pair of Linear Equations (Core-Basics Edition)

CHAPTER 3: PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

Ye page sirf un cheezon ka hai jo Class 7-8-9 mein padhayi jaati hain. Pura Chapter 3 inhi pe khada hai.

1) VARIABLE AUR EQUATION

Variable = unknown (x, y). Equation = jisme '=' ho.

Example: $2x + 3 = 7$ ek equation hai.

2) LINEAR EQUATION IN ONE VARIABLE

Sirf ek variable, power 1. Form: $ax + b = 0$.

$3x - 6 = 0 \rightarrow x = 2$. (Ek hi solution hota hai.)

3) LINEAR EQUATION IN TWO VARIABLES

Do variables, power 1. Form: $ax + by + c = 0$.

Example: $2x + 3y = 12$.

Iske INFINITE solutions hote hain (har x ke liye ek y).

4) SOLUTION KYA HAI?

Solution = (x, y) jo equation ko sahi (satisfy) kare.

$x+y=5$ ke liye (2,3) solution hai kyunki $2+3=5$.

5) GRAPH BASICS

X-axis horizontal, Y-axis vertical, milte hain ORIGIN

(0,0) pe. Point = (x, y).

Linear equation ka graph hamesha SEEDHI LINE hota hai.

6) RATIO (a1/a2 type)

Ratio = do numbers ki tulna (jaise $2/4 = 1/2$).

Is chapter mein coefficients ke ratio compare karenge:

$a1/a2$, $b1/b2$, $c1/c2$.

7) SIMPLE EQUATION SOLVE KARNA

Ek side ki cheez doosri side le jao to sign badalta:

$2x = 18 \rightarrow x = 18/2 = 9$. (divide karke nikalo.)

NOW: ASLI CHAPTER 3 SHURU

TOPIC 1: PAIR OF LINEAR EQUATIONS

Do linear equations ek saath, same variables:

$$a_1 x + b_1 y + c_1 = 0$$

$$a_2 x + b_2 y + c_2 = 0$$

SOLUTION = woh (x, y) jo DONO ko satisfy kare.

TOPIC 2: GRAPHICAL - 3 CASES

Dono lines ka graph banao, 3 possibilities:

- INTERSECTING (ek point pe milti) -> 1 solution
(Consistent)
- PARALLEL (kabhi nahi milti) -> 0 solution
(Inconsistent)
- COINCIDENT (ek hi line) -> infinite
(Consistent / dependent)

TOPIC 3: CONDITIONS BY RATIOS (bina graph ke)

EXAM ALERT: Ye table ratta maar - har exam mein aata hai.

$a_1/a_2 \neq b_1/b_2$ -> Unique solution (intersecting)

$a_1/a_2 = b_1/b_2 \neq c_1/c_2$ -> No solution (parallel)

$a_1/a_2 = b_1/b_2 = c_1/c_2$ -> Infinite (coincident)

Yaad: sab alag -> 1, do same ek alag -> 0, sab same -> infinite.

TOPIC 4: ALGEBRAIC METHODS

EXAM ALERT: Substitution aur Elimination se solve karna pakka aata hai (3 marks). Dono aane chahiye.

SUBSTITUTION:

1. Ek equation se ek variable doosre ke terms mein nikaalo
2. Doosri equation mein daal do
3. Single variable solve karo, phir wapas daalo

ELIMINATION:

1. Coefficients match karo (multiply karke)
2. Add/subtract karke ek variable gayab karo

3. Solve karo, phir wapas daalo

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples पूरी तरह solve karke. Upar HARDEST, neeche EASIEST. Step-by-step samajh.

Solved Example 1 (HARDEST) - Fraction word problem

(Exam favourite)

Ek fraction $\frac{9}{11}$ ban jaata hai agar numerator aur denominator dono mein 2 add karein. 3 add karein toh $\frac{5}{6}$ ban jaata hai. Fraction nikaalo.

Let fraction = $\frac{x}{y}$.

$$\frac{(x+2)}{(y+2)} = \frac{9}{11} \rightarrow 11(x+2)=9(y+2)$$

$$\rightarrow 11x - 9y = -4 \quad \dots(1)$$

$$\frac{(x+3)}{(y+3)} = \frac{5}{6} \rightarrow 6(x+3)=5(y+3)$$

$$\rightarrow 6x - 5y = -3 \quad \dots(2)$$

$$(1) \times 5: 55x - 45y = -20$$

$$(2) \times 9: 54x - 45y = -27$$

$$\text{Subtract: } x = 7$$

$$(2) \text{ mein: } 6(7) - 5y = -3 \rightarrow 42 + 3 = 5y \rightarrow y = 9$$

Fraction = $\frac{7}{9}$.

Solved Example 2 - Age word problem

(Exam favourite)

5 saal pehle Nuri, Sonu se 3 guna umar ki thi. 10 saal baad Nuri, Sonu se 2 guna hogi. Dono ki umar?

Let Nuri = x , Sonu = y .

$$x - 5 = 3(y - 5) \rightarrow x - 3y = -10 \quad \dots(1)$$

$$x + 10 = 2(y + 10) \rightarrow x - 2y = 10 \quad \dots(2)$$

$$(1) - (2): -y = -20 \rightarrow y = 20$$

$$(2) \text{ mein: } x - 40 = 10 \rightarrow x = 50$$

Nuri 50 saal, Sonu 20 saal.

Solved Example 3 - Elimination method

$$2x + 3y = 12 \quad \dots(1)$$

$$3x - 2y = 5 \quad \dots(2)$$

(1)x2: $4x + 6y = 24$

(2)x3: $9x - 6y = 15$

Add: $13x = 39 \rightarrow x = 3$

(1) mein: $2(3) + 3y = 12 \rightarrow 3y = 6 \rightarrow y = 2$

Solution: $x = 3, y = 2$.

Solved Example 4 - Substitution method

$x + 2y = 7 \dots(1)$

$3x - y = 0 \dots(2)$

(2) se: $y = 3x$

(1) mein daal: $x + 2(3x) = 7 \rightarrow 7x = 7 \rightarrow x = 1$

$y = 3(1) = 3$

Solution: $x = 1, y = 3$.

Solved Example 5 - Conditions (kaunsa case)

(Exam favourite - ratios se decide)

(a) $x+2y=5$ & $2x+4y=10$:

$1/2 = 2/4 = 5/10$ (sab same) $\rightarrow$ INFINITE (coincident)

(b) $2x+y=6$ & $4x+2y=7$:

$2/4 = 1/2$, par $6/7$ alag $\rightarrow$ NO solution (parallel)

(c) $3x-y=4$ & $x+2y=1$:

$3/1 \neq -1/2 \rightarrow$ UNIQUE solution (intersecting)

Solved Example 6 - Find k (infinite solutions)

(Exam favourite)

Kis k ke liye $2x+3y=7$ aur $(k-1)x+(k+2)y=3k$ ke INFINITE solutions honge?

Condition: $a_1/a_2 = b_1/b_2 = c_1/c_2$

$2/(k-1) = 3/(k+2) = 7/(3k)$

$2/(k-1) = 3/(k+2)$:

$2(k+2) = 3(k-1) \rightarrow 2k+4 = 3k-3 \rightarrow k = 7$

Check $7/(3k)$: $k=7 \rightarrow 2/6 = 1/3$ aur $7/21 = 1/3$ (match!)

So $k = 7$.

Solved Example 7 - Simple word (sum & difference)

Do numbers ka sum 50, difference 10. Numbers?

$$x + y = 50 \quad \dots(1)$$

$$x - y = 10 \quad \dots(2)$$

Add: $2x = 60 \rightarrow x = 30$; $y = 20$.

Solved Example 8 (EASIEST) - Solution check

(a) Kya (2,3), $2x+y=7$ ka solution hai?

$2(2)+3 = 7 \rightarrow$ haan, satisfy karta hai. Solution hai.

(b) $x+y=4$ ka ek solution: $x=1, y=3$ (kyunki $1+3=4$).

(Aise infinite solutions ho sakte hain.)

Q AND A TIME

Q1. CORE: $x+y=6$ ke 3 solutions likho.

Q2. Bina solve kiye batao (unique/no/infinite):

(a) $x+2y=4$ & $2x+4y=12$

(b) $3x-y=2$ & $6x-2y=4$

(c) $2x+y=5$ & $x-y=1$

Q3. Substitution se solve: $x+y=14$, $x-y=4$.

Q4. Elimination se solve: $3x+4y=10$, $2x-2y=2$.

Q5. Do numbers ka sum 25, difference 5. Numbers nikaal.

Q6. Kis k ke liye $kx+2y=5$ aur $3x+y=1$ ka UNIQUE solution hoga? (Hint: $a_1/a_2 \neq b_1/b_2$)

Q7. CORE CHECK: linear equation in 2 variables ke kitne solutions hote hain?

SUMMARY

1. Pair of linear equations $\rightarrow$ common (x,y) dhoondhna.

2. 3 cases: intersecting (1), parallel (0), coincident (infinite).

3. Ratios se:

$$a_1/a_2 \neq b_1/b_2 \quad \rightarrow \text{unique}$$

$$a_1/a_2 = b_1/b_2 \neq c_1/c_2 \quad \rightarrow \text{none}$$

$$a_1/a_2 = b_1/b_2 = c_1/c_2 \quad \rightarrow \text{infinite}$$

4. Substitution: ek variable replace karo.

5. Elimination: add/subtract karke ek variable gayab.

CORE (RED page) revise: solution ka matlab, graph =
seedhi line, ratio compare karna.

Generated by Kiro for Krishna1k

Class 10 NCERT Maths - Chapter 3: Pair of Linear Equations (Core-Basics edition)
